Ministry of Education
Umm Al-Qura University
Collage of computer and information system

وزارة التعليم
جامعة ام القرى
كلية الحاسب الالي ونظم المعلومات

Book Store Website Project

Phase 1

Submitted by:

Shaima Mohammed Al-Khuzaee - [redacted]

Hala Mohammed Jalal - [redacted]

Razan Mohammed Hakimi - [redacted]

Lujain Khaled Al-Sayed – [redacted]

Shahd Yousef Almatrouk-[redacted]

Course:

Advanced Web Programming

Table of content

Table of content 2

1.1 Project Overview 4

2. System Design and Development 5

2.1 Database Design and ER Diagram 5

2.1.1 Data Model 5

2.1.2 ER Diagram 7

2.2 User Roles and Permissions 9

2.2.1 User Roles 9

2.3 Interface Prototype 11

2.3.1 Wireframes 11

2.3.2 Mockups 13

2.3.3 User Flows & Journey 17

2.4 Backend Development 19

2.5 Implementation Details ..... **Error! Bookmark not defined.**

3. Testing and Quality Assurance 20

3.1 Testing Plan 20

3.1.1 Target User Group 20

3.1.2 Testing Scenarios 20

3.1.3 Testing Methods 22

3.1.4 Data Analysis Plan 22

3.1.5 Ethical Considerations 23

4. Human-Computer Interaction (HCI) & Error Handling Enhancements 24

4.1 Usability Enhancements 24

4.2 Error Handling Mechanisms 25

4.3.1 Graceful Degradation 25

4.3.2 Informative Error Messages 25

4.3.3 Preventive Measures 25

4.3.4 Logging & Monitoring 26
5. Future Plans 27
6. Conclusion.....30
7. References 31

1. Introduction

The **Book Store Website** project aims to create an engaging and user-friendly online platform for exploring books and their categories. Designed to cater to a wide range of users, from casual visitors to dedicated book enthusiasts, the website allows users to browse book listings, view detailed author information, and interact with the content through reviews and ratings.

Registered users benefit from personalized experiences, including account management, the ability to leave reviews, and access to external stores for book purchases. The admin role enhances the platform's functionality by managing content, moderating reviews, and ensuring smooth operation. This project not only simplifies book exploration but also bridges the gap between users and popular online retailers, creating a seamless journey for book lovers.

1.1 Project Overview

The Book Store Website is designed to provide an interactive and user-friendly experience for book lovers. The website allows users to explore various book categories, view detailed information on books and authors, and interact through ratings and reviews. Users can register for personalized experiences, including managing their favorite books and leaving reviews.

The platform also integrates external book retailers, allowing users to purchase books seamlessly. The system differentiates between users (customers), employees, and administrators, each with distinct roles and permissions. The website is developed with a structured database, a secure backend, and an intuitive frontend interface.

2. System Design and Development

2.1 Database Design and ER Diagram

2.1.1 Data Model

1. **1Users / Account Table**
    - UserID (Primary Key)
    - Username (Unique)
    - Password
    - Email (Unique)
    - Role
    - RegistrationDate
    - list item

2. **reviews Table**
    - ReviewID (Primary Key)
    - UserID (Foreign Key)
    - BookID (Foreign Key)
    - Rating
    - ReviewText
    - ReviewDate

3. **Favorite Table**
    - SavedID (Primary Key)
    - UserID (Foreign Key)
    - BookID (Foreign Key)

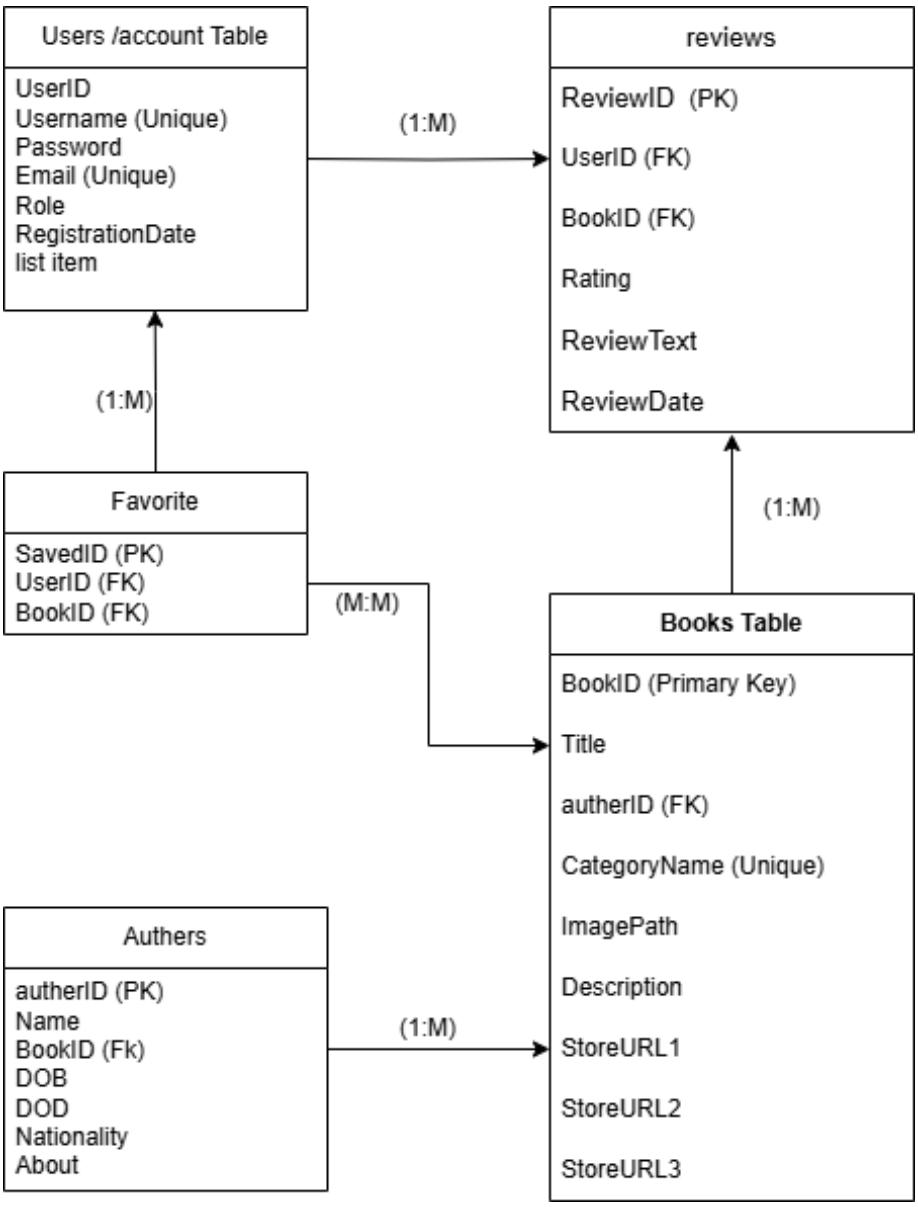
4. **Authors Table**
    - authorID (Primary Key)
    - Name
    - BookID (Foreign Key)
    - DOB
    - DOD
    - Nationality
    - About

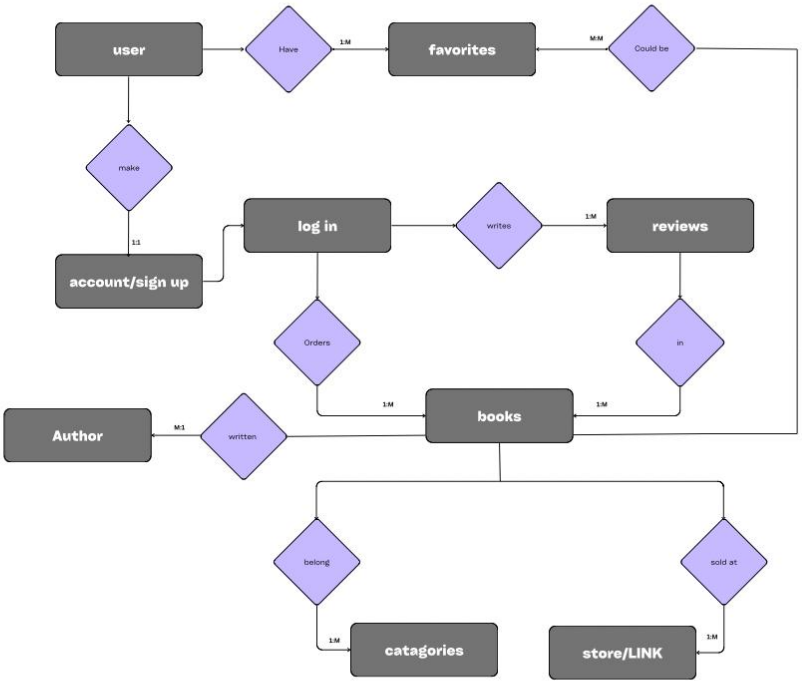
5. **Books Table**
    - BookID (Primary Key)
    - Title
    - authorID (Foreign Key)
    - CategoryName (Unique)
    - ImagePath
    - Description
    - StoreURL1
    - StoreURL2
    - StoreURL3

Relationships:

- **Users to reviews: One-to-Many (1:M)**
- **Users to Favorite: Many-to-One (M:1)**
- **Favorite to Books: Many-to-One (M:1)**
- **reviews to Books: One-to-Many (1:M)**
- **Books to Authors: One-to-Many (1:M)**

2.1.2 ER Diagram




user
Have
1:M
favorites
M:M
Could be
make
1:1
account/sign up
log in
writes
1:M
reviews
Orders
in
1:M
1:M
books
Author
M:1
written
belong
sold at
1:M
catagories
store/LINK
1:M

2.2 User Roles and Permissions

2.2.1 User Roles

1. User (Customer):

   Privileges:
   - Can view book details.
   - Can write reviews.
   - Can access the “Buy Now” feature to purchase books.

   Limitations:
   - Cannot manage user accounts.
   - Cannot add, edit, or delete books.
   - Cannot moderate reviews.

2. Employee:

   Privileges:
   - Can add and edit book details.
   - Can moderate reviews.

   Limitations:
   - Cannot delete books.
   - Cannot manage user accounts.

3. Admin:

   Privileges:
   - Can manage user accounts (add, edit, delete).
   - Can add, edit, and delete books.
   - Can moderate reviews.
   - Can view all data, including reports and analytics.

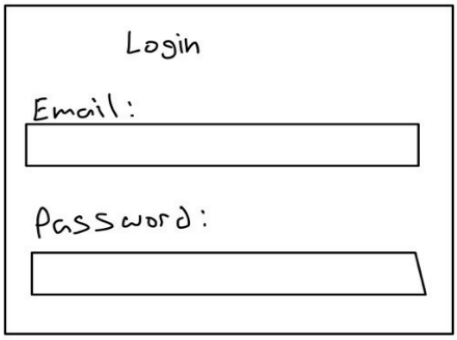
   Limitations:
   - Cannot write reviews.
   - Cannot access “Buy Now” feature

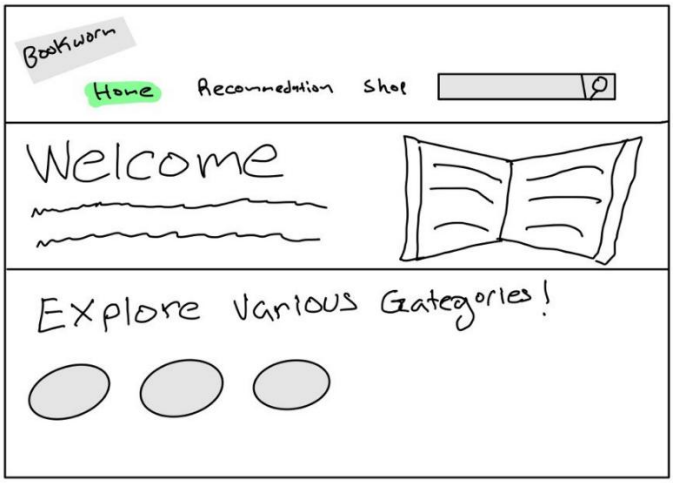
2.2.2 Permission Matrix

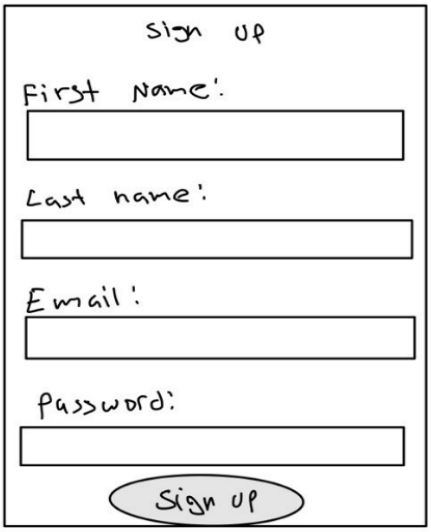
Functionality/Resource	Registered User (Customer)	Admin	Employee
View Book Details	Yes	Yes	Yes
Write Reviews	Yes	No	No
Access "Buy Now" to Purchase Books	Yes	No	No
Edit Personal Account Info	Yes	Yes	No
Add Books	No	Yes	Yes
Edit Book	No	Yes	Yes
Delete Book	No	Yes	No
Moderate Reviews	No	Yes	Yes
Manage users accounts	No	No	Yes
View reports/analytics	No	No	Yes

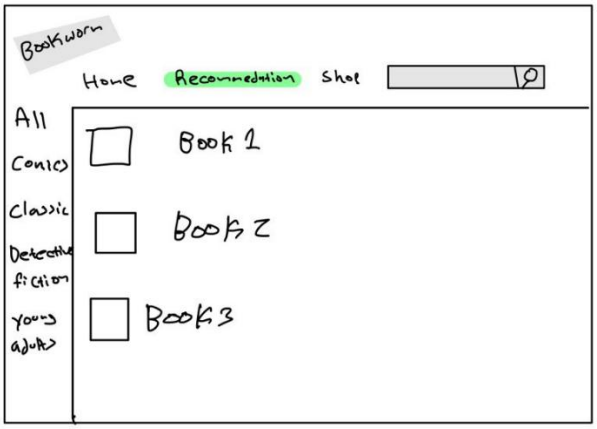
2.3 Interface Prototype

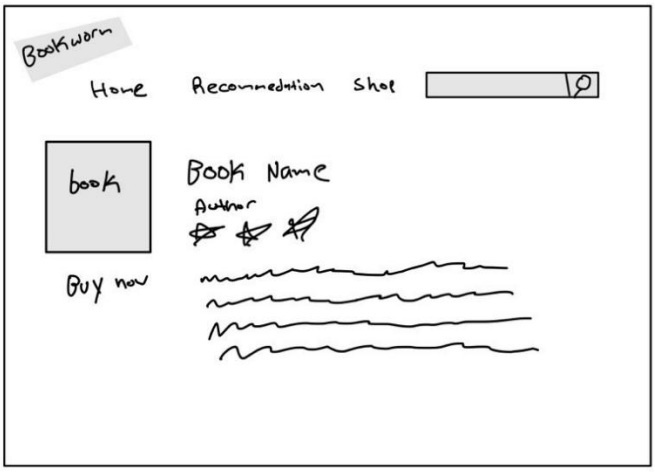
2.3.1 Wireframes

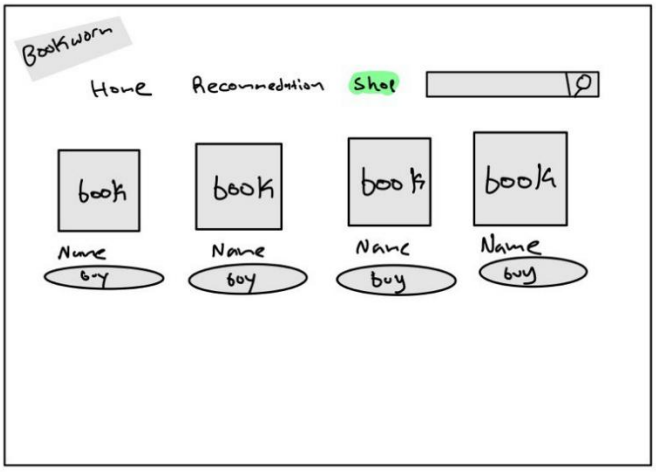


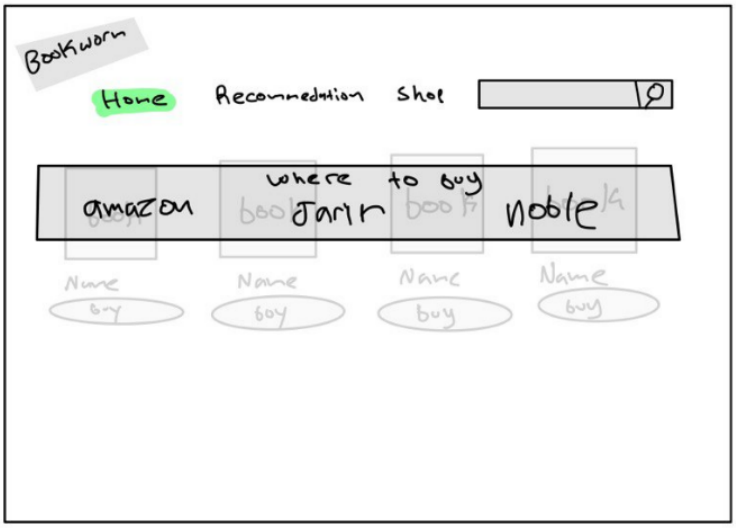




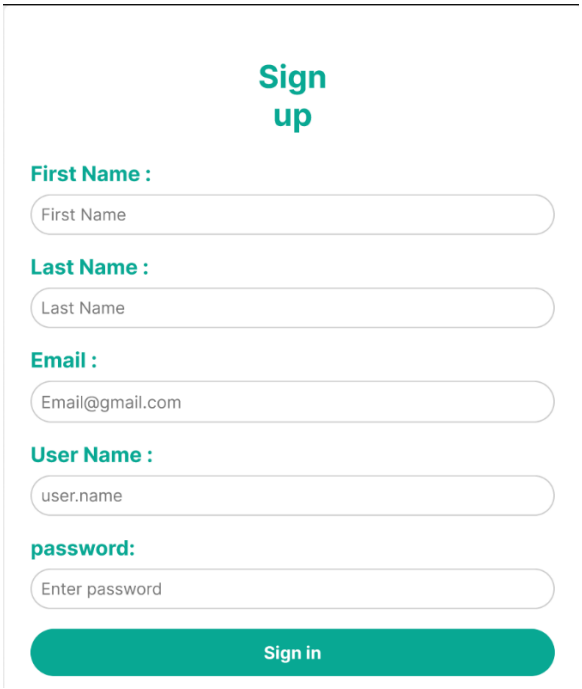


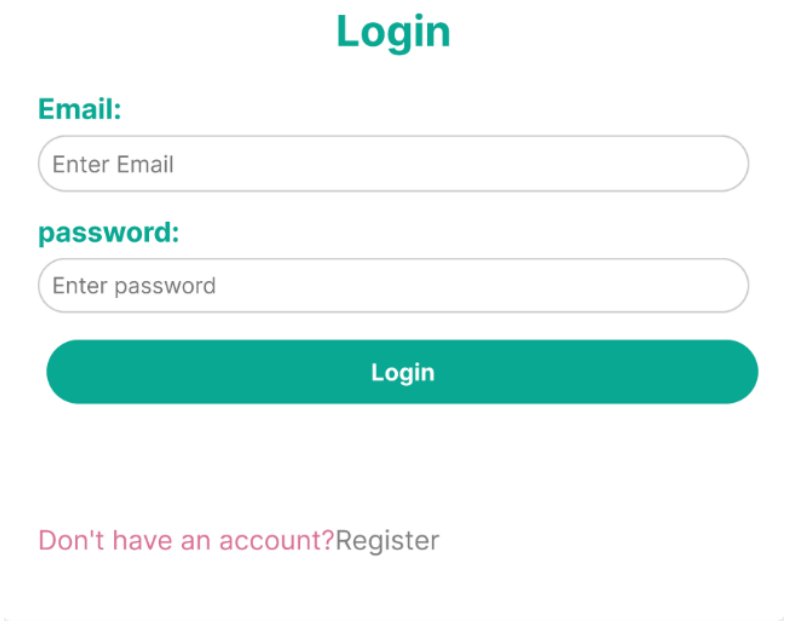

Bookworm
Home
Recommendation
Shop
book
Book Name
Author
Buy now


Bookworm
Home
Recommendation
Shop
book
book
book
book
Name
Name
Name
Name
buy
buy
buy
buy


Bookworm
Home
Recommendation
Shop
where to buy
amazon
barn
noble
Name
Name
Name
Name
buy
buy
buy
buy

2.3.2 Mockups


Sign up
First Name :
First Name
Last Name :
Last Name
Email :
Email@gmail.com
User Name :
user.name
password:
Enter password
Sign in


Login
Email:
Enter Email
password:
Enter password
Login
Don't have an account?Register

BOOKWORM

Home Recommedations Shop Search books

Welcome

To where the magic of words *wriggles* off the page and into your soul

Like worms tunneling through the earth, we explore the vast expanse of literature unearthing hidden treasures and sharing tales that wiggle their way into your imagination. So, grab your literary shovel and dig in with us at Bookworm, where every book is a journey and every reader a cherished companion in our wormy wonderland.



Explore Various Categories!







BOOKWORM

Home Recommedations Shop Search books Sign up Log in

Rating & Reviews

Write a Review

What do you think?

Layn2005
1 day ago
★★★★
is book made me laugh, cry, and feel a lot emotions. I can proudly say that this is the EST book I have ever read

AboHussn
3 days ago
★★★★½
This fiction novel is told from the perspective of a ten-year-old boy named August, Auggie, who was born with a facial deformity. He bravely faces the challenges of fitting in at a new school and tries to convince his new classmates that he is just like them despite his appearance.

Candance
1 month ago
★★★★★
Wow! I am so glad that I finally got around to listening to this story. 'Wonder' had been sitting on my TBR list for a really long time, but I had been waiting for a time to listen to it with my daughters. That time finally arrived last weekend, when we had to spend a full day in the car on a trip.

Lana.Jobs
2 months ago
★★★★½
Incredibly wholesome. A book that makes you cry but is also uplifting throughout. Should be a compulsory text for school kids."

Need Help?

Connect Our Support Team

BOOKWORM

Home Recommedations Shop

Search books

All

Comics

Horror

Classic

Detective Fiction

Young Adults



The Number One Young Adults #1

Alice's Adventures in Wonderland

Author : Lewis Carroll



The Number Two Young Adult #2

Wonder

Author : R.J. Palacio



The Number Three Young Adult #3

Charlotte's Web

Author : Elwyn Brooks White



The Number One Detective Fiction #1

Black coffee

Agatha Christie

BOOKWORM

Home Recommedations Shop

Search books



Buy Now

The Number One Young Adults #1

Alice's Adventures in Wonderland

Lewis Carroll

★★★★★ 5.00

Category: Young Adults

Would you tell me, please, which way I ought to go from here?

Alice's Adventures in Wonderland is an 1865 novel written by English author Charles Lutwidge Dodgson under the pseudonym Lewis Carroll. It tells of a girl named Alice falling through a rabbit hole into a fantasy world populated by peculiar, anthropomorphic creatures. The tale plays with logic, giving the story lasting popularity with adults as well as with children. It is considered to be one of the best examples of the literary nonsense genre. Its narrative course and structure, characters and imagery have been enormously influential in both popular culture and literature, especially in the fantasy genre.

Language: English 70 pages

☆☆☆☆☆

Review title

Review body

Reviewer name

Date

BOOKWORM

Home Recommedations Shop

Search books



Alice's Adventures in Wonderland

Buy Now



As Good As Dead

Buy Now



Batman: Year One

Buy Now

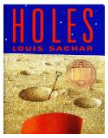


Kawf (كوف)

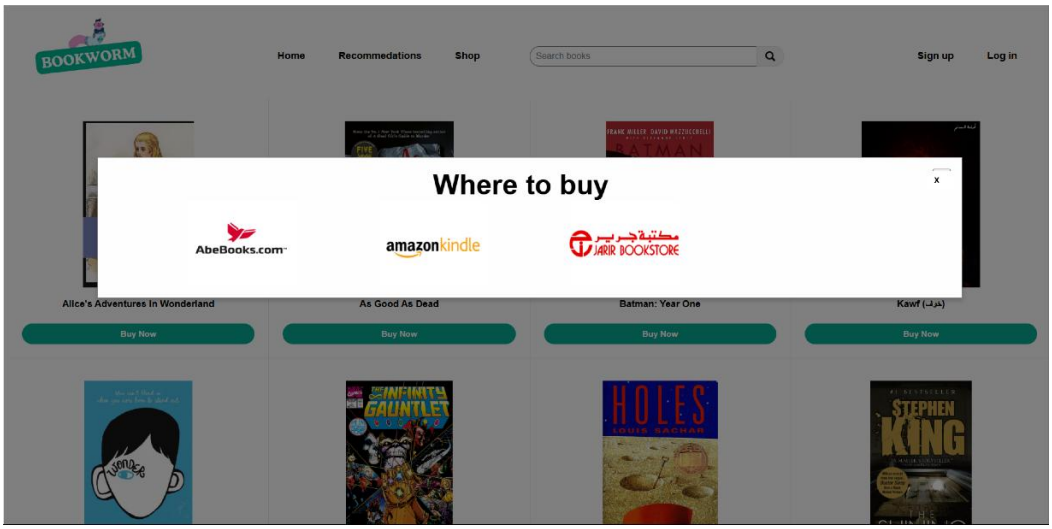
Buy Now










BOOKWORM
Home
Recommedations
Shop
Search books
Sign up
Log in
Where to buy
AbeBooks.com
amazonkindle
مكتبة جرير
JARIR BOOKSTORE
Alice's Adventures In Wonderland
As Good As Dead
Batman: Year One
Kawf (كرف)
Buy Now
Buy Now
Buy Now
Buy Now
INFINITY GAUNTLET
HOLES
LOUIS SACHAR
STEPHEN KING

2.3.3 User Flows & Journey

User Journey for Bookworm:

1. **Account Creation and Login:**
    a. The user opens the Bookworm application.
    b. On the homepage, the user clicks on the **"Sign Up"** button.
    c. They fill in their details (name, email, password) and submit the form.
    d. A confirmation message appears, and the user logs in using their credentials.
2. **Navigating to Recommendations:**
    a. After logging in, the user is taken to their personalized dashboard.
    b. From the dashboard, the user clicks on the **"Recommendations"** section in the navigation bar.
    c. The Recommendations page displays categories: **Comics**, **Horror**, **Classic**, **Detective Fiction**, and **Young Adults.**
3. **Exploring Categories and Books:**
    a. The user selects a category (e.g., **Comics**), which opens a list of books in that category.
    b. The user scrolls through the list and clicks on a specific book to view its details.
4. **Reading and Writing Reviews:**
    a. On the book details page, the user sees:
        i. Book title, author, description, and cover.
        ii. Average rating and user reviews.
    b. The user clicks **"Write a Review"**, enters their review, and submits it.
    c. The new review appears in the list of reviews.
5. **Purchasing a Book:**
    a. The user clicks on the **"Buy Now"** button for the book.
    b. They are taken to the **Shop** page, where the book is listed.
    c. The user clicks on the book, which redirects them to a store (e.g., Amazon, Barnes & Noble, Jarir) that sells the book.
6. **Store Redirection:**
    a. The user completes their purchase on the chosen store's website.

This journey ensures a smooth experience from account creation to book exploration and purchasing while allowing user engagement through reviews.

Pain Points and Opportunities:

1. Account Creation and Login:

- **Pain Points:**
  - Users may feel frustrated if the signup process is too lengthy or requires unnecessary information.
  - Forgotten passwords can disrupt the user's experience.
- **Opportunities for Improvement:**
  - Simplify the signup form with only essential fields (e.g., email, password).
  - Implement a clear and user-friendly "Forgot Password" feature.

2.4 Backend Development

User Management: Handling user registration, login processes, role assignment, and session management. The backend is responsible for securely storing and managing user data, including usernames, passwords, email addresses, and roles.

Book Management: Adding, updating, and deleting book entries in the database. This includes managing all attributes associated with books such as titles, authors, categories, descriptions, and store URLs.

Author Management: Adding and maintaining author details in the database, including their biography, books written, and any relevant personal details like date of birth or nationality.

Comments and Ratings: Processing and storing user comments and ratings on books. This includes linking comments to specific users and books, managing the submission date of comments, and updating ratings.

Favorites System: Managing the user's favorite books. This includes adding and removing books from a user's favorites list and ensuring that these changes are reflected in the database. The system utilizes XML for structured storage and JSON for API communication, ensuring efficient data exchange and user experience.

Access Control and Permissions: Enforcing different levels of access and permissions for different types of users. For example, regular users can view books and write reviews, whereas admins have the capabilities to edit or delete book entries and manage user accounts.

Integration with External Book Stores: Handling the redirections to external book stores (e.g., Amazon, Barnes & Noble) from the book detail pages. This typically involves constructing URLs dynamically based on the book data and managing affiliate links if applicable.

Session and State Management: Maintaining the state of user sessions, ensuring that users are logged in and their session data is consistent as they navigate through different parts of the website.

Data Validation and Security: Ensuring that all data entering the system via user inputs (like comments, new book details, or user registration information) is properly validated and sanitized to prevent SQL injection, XSS attacks, and other security threats.

3. Testing and Quality Assurance

3.1 Testing Plan

3.1.1 Target User Group

Demographics:

- Age: 18-45 years old
- Location: Global (with emphasis on English-speaking users)
- Gender: All genders
- Occupation: Students, young professionals, book enthusiasts
- Technical Skills: Basic to intermediate (familiar with navigating websites and using apps)

Needs:

- Easy access to book recommendations and reviews.
- Streamlined account creation and login process.
- Smooth and intuitive navigation between categories and book details.
- Ability to read and write reviews.
- Easy book purchasing flow with redirection to external store

3.1.2 Testing Scenarios

1- Account Creation and Login

- Scenario: A user is asked to create an account and log in.
- Task:
    - Open the Bookworm application.

- Sign up with a name, email, and password.
- Log in with the created credentials.

- Objective: Test the ease of use of the sign-up form and login process.

2- Navigating to Favorites

• Scenario: A user is asked to navigate to the "Favorites" section.

• Tasks:

- After logging in, access the dashboard.
- Click on "Favorites" in the navigation bar.
- View and manage the favorite books list.

• Objective:

- Test how intuitive and easy it is for users to navigate through the favorites section.
- Ensure users can effortlessly access and manage their favorite books.

3- Exploring Categories and Books

- Scenario: A user explores a category and views book details.
- Task:
  - Select a category (e.g., Comics).
  - Click on a book to view its details (title, author, description, reviews).
- Objective: Ensure users can easily explore categories and find detailed information on books.

4- Purchasing a Book

- Scenario: A user purchases a book.
- Task:
  - Click the "Buy Now" button for a book.
  - Navigate to the Shop page and proceed to the external store (e.g., Amazon).

- Objective: Test the effectiveness and smoothness of the purchasing flow, including redirection to external store

3.1.3 Testing Methods

For the evaluation of the Book Store Website, a **questionnaire** will be used to collect user feedback. The questionnaire aims to gather insights into various aspects such as usability, functionality, design, and overall user experience. Participants will provide their responses via a Google Form, which can be accessed through the following link: [https://forms.gle/kX5RFQQQZMAgM9dk7].

3.1.4 Data Analysis Plan

The data collected from the questionnaire will be analyzed using quantitative methods:

1. Statistical Analysis:
    - Responses will be aggregated and analyzed using statistical methods such as percentages, mean scores, and frequency distributions to identify patterns and trends.
    - Key usability metrics such as task completion rates, user satisfaction scores, and response time data will be evaluated.

3.1.5 Ethical Considerations

To ensure user privacy and data security during the testing process, the following measures will be implemented:

1. **Informed Consent:**
    - Participants will be informed about the purpose of the study, data usage, and their rights before filling out the questionnaire.
    - They will have the option to withdraw at any time without consequences.
2. **Anonymity and Confidentiality:**
    - Personal identifiers will not be collected, ensuring anonymity of responses.
    - Data will be stored securely with restricted access to authorized personnel only.
3. **Data Security:**
    - Collected data will be encrypted and stored in compliance with data protection regulations to prevent unauthorized access or breaches.
    - Google Form settings will be configured to limit access and prevent data sharing outside the research team.
4. **Compliance with Ethical Standards:**
    - The study will adhere to relevant ethical guidelines and university policies to maintain high ethical standards in data collection and analysis.

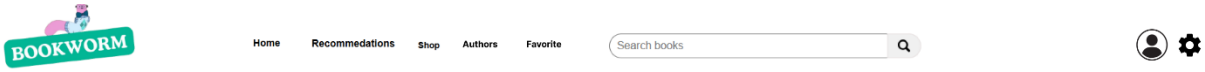
4. Human-Computer Interaction (HCI) & Error Handling Enhancements (lab7)

4.1 Usability Enhancements

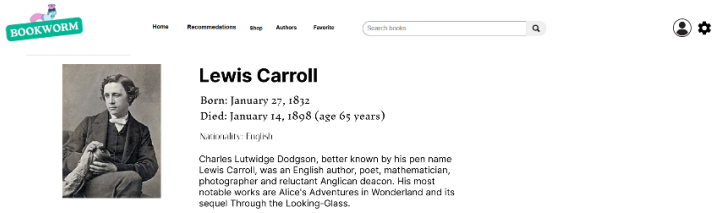
- To enhance accessibility for individuals with disabilities, we will implement both light and dark mode options on the website, allowing users to select their preferred mode.

- A warning message will be displayed if the user:
  - clicks "Sign In" without filling in the fields, or
  - enters input that does not match the database.

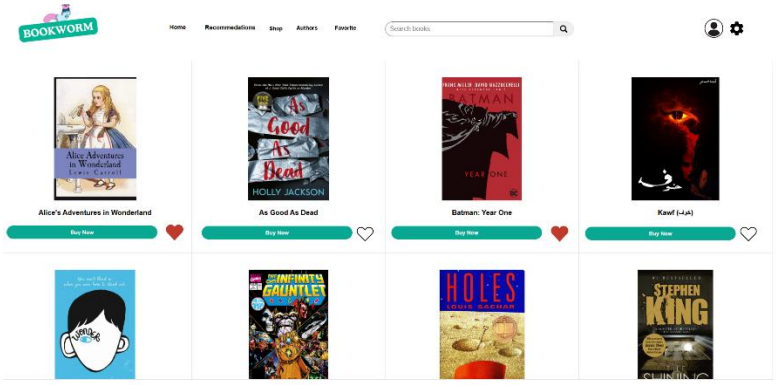
- A success message will be displayed when the user successfully creates an account during the sign-in process.



- Modified the menu bar for better accessibility



- Added an author page



- Added a favorite button so the user can come bake to the books he wants to buy.

4.2 Error Handling Mechanisms

4.3.1 Graceful Degradation

To ensure the website operates smoothly even during unexpected failures, we have implemented mechanisms to handle:

Network Errors:

Solution: Display a user-friendly message, e.g., “Unable to connect to the external store. Please try again later.”

Server Downtime:

Solution: Show an informative alert like “The server is currently unavailable. Please try again shortly.”

Invalid User Input:

Solution: Guide users with real-time error prompts to correct their input.

4.3.2 Informative Error Messages

Providing clear and concise error messages to guide users, such as:

- Invalid email input: “Please enter a valid email format (e.g., example@domain.com).”
- Empty fields: “All fields are required. Please fill in the missing details.”
- Login failure: “Incorrect username or password. Please try again.”

4.3.3 Preventive Measures

To minimize common errors, we have implemented the following strategies:

Input Validation:

- Ensuring valid email format during registration.
- Password must meet security criteria (e.g., 8+ characters, special characters).
- Reviews must have a valid rating between 1-5 stars.

Data Type Checking:

- Numeric fields like phone numbers cannot contain letters.
- Image uploads restricted to accepted formats (JPEG, PNG).

Boundary Checks:

- Limiting the number of characters for text fields (e.g., book titles, reviews).
- Setting maximum book quantities in the shopping cart.

4.3.4 Logging & Monitoring

To track errors and potential issues, we have implemented:

Log Files:

- Capturing timestamps, error types, and user actions.
- Example: Logging failed login attempts for security purposes.

Monitoring Tools:

- Using analytics tools like Google Analytics to monitor user interactions and identify pain points

5. Future Plans

1. **New Features:**

Objective: Improve the browsing experience and make book exploration more seamless and user-friendly.

Proposed Enhancements:

- **Smart Search Optimization:**

  Implement Live Search to display suggestions as users type.

  Add advanced filters (e.g., category, author, price, rating, language).

  Support automatic spelling correction in search queries.

- **Enhanced Book Browsing System:**

  Provide Grid View and List View options for displaying books conveniently.

  Introduce a Quick Preview Mode to allow users to read a book summary before visiting the store.

  Book Comparison Feature:

  Allow users to compare multiple books to evaluate differences before purchasing.

- **User-Generated Comments:** Users will be able to **write and reply to comments** on book pages, adding a more interactive community feel. Challenges like **comment moderation, preventing spam, and ensuring a seamless UI experience** will be tackled to provide a secure and efficient commenting system.

2. **Scalability**

Objective: Optimize the website's performance to handle increased traffic efficiently.

Proposed Enhancements:

Database Optimization:

Implement Indexing and Caching to improve search and book display performance**.**

Content Delivery Network (CDN) Integration:

Utilize Cloudflare or AWS CloudFront to speed up content loading worldwide.

Optimized Data Loading:

Apply Lazy Loading to ensure that images and content load only when needed.

Load Balancing Implementation:

Distribute user traffic across multiple servers to ensure smooth and responsive performance.

3. Integration with Third-Party Services

Objective: Expand the website's capabilities by integrating with external services.

Proposed Enhancements:

Social Media Integration:

Enable social login options via Google, Facebook, and Twitter.

Allow users to share book reviews on social media platforms.

Improved External Store Links:

Display multiple store options per book, including pricing comparisons (e.g., Jarir – $50, Amazon – $45).

Add a prominent "Buy Now" button to streamline the transition to external stores.

External Review System Integration:

Display book ratings and reviews from platforms like Goodreads and Amazon directly on the website.

4. Artificial Intelligence & Machine Learning

Objective: Enhance user experience by implementing AI-driven recommendations and analytics.

Proposed Enhancements:

Personalized Book Recommendations:

Utilize AI to suggest books based on user interests and browsing history.

Highlight trending books based on popular choices over a specific period.

User Behavior Analysis:

Track visitor behavior to understand which books are most browsed and generate smart recommendations accordingly.

AI Chatbot Assistance:

Implement a chatbot to help users find books based on their queries and preferences.

5. Security & Data Protection

Objective: Secure user data and prevent fraudulent activities.

Proposed Enhancements:

Data Encryption:

Use SSL/TLS encryption to ensure that all communications between the user and the server are secure.

Broken Link Detection:

Implement an automated system to regularly check and fix broken external store links.

Spam & Fake Review Prevention:

Use reCAPTCHA to prevent spam accounts.

Deploy AI-based review analysis to detect and filter out fake reviews.

6. Enhancing the Experience for Registered Users

Objective: Encourage users to register and increase engagement on the platform.

Proposed Enhancements:

Favorite Books Section:

Allow users to save books to a wishlist for future reference.

Personalized Dashboard:

Display browsing and search history, allowing users to revisit previously viewed books.

Custom Notifications:

Send alerts for book price drops, availability updates, and personalized recommendations.

6. Conclusion

The Book Store Website project is an initiative to create an intuitive, feature-rich online bookstore that caters to book enthusiasts. By incorporating interactive elements such as user reviews, favorite book tracking, and seamless purchasing options, the website aims to enhance the reading experience. The structured design of the system ensures security, usability, and scalability, making it adaptable for future enhancements, such as AI-powered recommendations and social media integration.

With planned usability improvements and security enhancements, this project lays the foundation for an engaging and efficient online book store that connects users with their favorite books effortlessly.

7. References

- Krug, S. (2014). Don't Make Me Think, Revisited: A Common Sense Approach to Web Usability. New Riders.
- Ferraiolo, D., Kuhn, R., & Chandramouli, R. (2003). Role-Based Access Control. Artech House.
- Flanagan, D. (2020). JavaScript: The Definitive Guide. O'Reilly Media.
- Norman, D. (2013). The Design of Everyday Things: Revised and Expanded Edition. Basic Books.
- Amazon Developer Guide: https://developer.amazon.com/docs/Barnes & Noble Partner Portal: https://press.barnesandnoble.com/ Jarir API Documentation: https://www.jarir.com/en/help